A Comprehensive Measure of Well Being

Project Description:

HDI Predictor — Policy Reviewer harnesses real-time computation and Generative AI to provide instant, data-driven analysis of a country's Human Development Index (HDI). The platform includes a Human Capital Predictor module, which computes the composite HDI score from four core indicators using the UNDP geometric mean formula; a Policy Intervention Simulator, which projects the impact of real-world policy actions on a country's baseline development trajectory; and a Global Comparison Dashboard, which benchmarks any custom country profile against real nations and global averages across all three HDI dimensions.

Utilizing Google's Gemini API for the AI Policy Advisor feature, the platform processes a country's current indicator profile and HDI classification to generate a personalized policy advisory report with targeted administrative reform recommendations. Built with React, TypeScript, and an Express.js backend, the platform ensures a fast, type-safe, and responsive experience. With secure API key management via environment variables and structured indicator validation, HDI Predictor empowers policy analysts, government researchers, and development economists to explore human development scenarios with confidence.

Scenarios

Scenario 1: A policy analyst selects a country profile with high life expectancy (78 years), strong mean and expected years of schooling (12.5 / 17 years), and a high GNI per capita ($42,000 PPP). The system computes a Very High HDI score (≥0.800), highlighting the country's strong performance across all three development pillars and confirming its position among the most developed nations.

Scenario 2: A researcher inputs a custom profile representing a developing nation — moderate life expectancy (46.5 years), average schooling attainment (10.0 / 14.4 years), and a mid-range GNI per capita ($27,900 PPP). The tool returns a Medium HDI score of 0.6336, flags substantial longevity deficits as the key weakness via its Sovereign Diagnostics panel, and generates targeted health-sector policy recommendations with estimated HDI score boosts for each intervention.

Scenario 3: A government advisor uses the Policy Intervention Simulator to test the projected impact of a National Primary Healthcare Revitalization initiative (+0.035 estimated boost) combined with a Secondary Education Access program (+0.012 estimated boost). The platform displays the baseline versus simulated HDI side by side, allowing the advisor to justify budget allocation using quantified, evidence-based projections before presenting to policymakers.

Technical Architecture:

Description: HDI Predictor utilizes a modular client-server architecture to deliver instant, AI-enhanced development analysis. The frontend is a React + TypeScript single-page application providing a responsive slider-based interface, while a lightweight Express.js backend handles a single specialized endpoint for AI-driven policy advisory generation. The core HDI computation (geometric mean of health, education, and income sub-indices) runs entirely client-side for instant feedback, while the backend interfaces with Google's Gemini API to generate deeper, context-aware policy recommendations on demand.

Layer	Component	Responsibility
Frontend	React + TypeScript (App.tsx)	Manages application state, indicator inputs, scenario/preset loading, and routes between the Predictor, Comparison, Report, and About views
Computation	calculateHdiFromIndicators()	Normalizes each raw indicator to a 0-1 sub-index using UNDP goalposts, then computes the geometric mean composite HDI score
Simulation	getSimulatedValues()	Applies selected policy intervention boost multipliers to baseline indicators to project a simulated HDI outcome
Backend	Express server (server.ts)	Exposes POST /api/advisor, receives indicator values and HDI category, and forwards a structured prompt to the Gemini API
AI Integration	Google Gemini API	Generates a personalized policy advisory report with strengths, weaknesses, and targeted administrative reform recommendations

Note: This project does not use a persistent database — all reference data (country presets, policy interventions, global averages) is stored as static in-memory TypeScript objects in data.ts, so no ER diagram is required.

Pre-requisites:

- **React & TypeScript Knowledge:** React Documentation, TypeScript Handbook
- **Gemini API Familiarity:** Google AI Studio Documentation
- **Node.js & Express Proficiency:** Express.js Documentation
- **Vite Build Tool:** Vite Documentation
- **HTML, CSS, and JavaScript Skills:** W3Schools HTML/CSS/JavaScript Tutorials
- **Version Control with Git:** Git Documentation
- **UNDP HDI Methodology:** Human Development Report Technical Notes

Project Workflow:

Milestone 1: Model Definition and Architecture

- **Activity 1.1:** Research the UNDP HDI methodology and define the normalization goalposts for life expectancy, schooling, and GNI per capita.
- **Activity 1.2:** Select the appropriate computation approach — client-side geometric mean calculation for instant feedback versus server-side processing.
- **Activity 1.3:** Define the application architecture, detailing interactions between the frontend, the HDI computation engine, and the Gemini AI advisor.
- **Activity 1.4:** Set up the development environment, installing React, TypeScript, Vite, and Express dependencies.

Milestone 2: Core Functionalities Development

- **Activity 2.1:** Develop the core HDI Predictor: sub-index computation, composite score calculation, and tier classification.
- **Activity 2.2:** Implement the Policy Intervention Simulator with baseline versus simulated HDI comparison.

Milestone 3: Backend & AI Advisor Development

- **Activity 3.1:** Write the Express server logic in server.ts, establishing the /api/advisor route and integrating Gemini AI responses.

Milestone 4: Frontend Development

- **Activity 4.1:** Design and develop the user interface using React, TypeScript, and Tailwind-style utility classes, ensuring a responsive, dark-themed layout.
- **Activity 4.2:** Create dynamic components — HdiPredictor, GlobalComparison, HdiReport, AboutHdi, and AboutProject — with live state-driven rendering.

Milestone 5: Deployment

- **Activity 5.1:** Prepare the application for local deployment by configuring the Vite build and Express server environment.
- **Activity 5.2:** Deploy the application publicly to a static hosting platform, exposing the Express advisor endpoint via a serverless function or dedicated Node process.

Milestone 6: Conclusion

Milestone 1: Model Definition and Architecture

In this milestone, the focus is on defining the mathematical model behind the HDI Predictor and designing the overall system architecture. This involves researching the UNDP's official HDI methodology and translating the geometric mean formula into an efficient, real-time client-side computation, then extending it with an AI-powered advisory layer.

Activity 1.1: Define the UNDP Normalization Goalposts

Before any prediction can be made, each raw indicator must be normalized to a 0–1 sub-index using fixed minimum and maximum goalposts as defined by the UNDP Human Development Report technical notes.

```
// types.ts — core HDI computation types
export interface HdiComputation {
  lifeExpectancy: number; // years (goalpost: 20-85)
  meanYearsSchooling: number; // years (goalpost: 0-15)
  expectedYearsSchooling: number; // years (goalpost: 0-18)
  gniPerCapita: number; // PPP $ (goalpost: 100-75000, log scale)
  healthIndex: number;
  educationIndex: number;
  incomeIndex: number;
  hdiScore: number;
  category: 'Very High' | 'High' | 'Medium' | 'Low';
}
```

Activity 1.2: Select the Computation Approach

- **Client-side computation:** Chosen for the core HDI formula to deliver instant slider feedback with zero network latency.
- **Server-side AI advisory:** Chosen for the policy recommendation layer, since generating nuanced administrative reform suggestions requires a large language model (Gemini) rather than deterministic rules.

Activity 1.3: Define the Application Architecture

- **Draft an Architectural Diagram:** Represent the frontend (React components), computation engine, simulation layer, and Gemini AI integration.
- **Detail Frontend Functionality:** Outline how users adjust sliders, select country/scenario presets, toggle policy interventions, and request an AI-generated advisory report.
- **Outline Backend Responsibilities:** Specify how the Express server handles incoming POST requests to /api/advisor, validates the indicator payload, and constructs a structured prompt for Gemini.
- **Describe AI Integration Points:** Define how the backend builds a prompt combining the HDI category and sub-index values, calls the Gemini API, and returns a structured JSON advisory response to the frontend.

Milestone 2: Core Functionalities Development

Activity 2.1: Develop the HDI Computation Engine

Implement the core function that transforms four raw indicators into a composite HDI score and tier classification, following the UNDP geometric mean approach.

```
// App.tsx — core HDI calculation logic
function calculateHdiFromIndicators(
  lifeExpectancy: number,
  meanYears: number,
  expectedYears: number,
  gniPerCapita: number
): HdiComputation {
  const healthIndex = (lifeExpectancy - 20) / (85 - 20);
  const eduIndex = ((meanYears / 15) + (expectedYears / 18)) / 2;
  const incomeIndex = (Math.log(gniPerCapita) - Math.log(100)) /
    (Math.log(75000) - Math.log(100));
  const hdiScore = Math.cbrt(healthIndex * eduIndex * incomeIndex);
  const category = classifyTier(hdiScore);
  return { healthIndex, educationIndex: eduIndex, incomeIndex, hdiScore, category };
}
```

Activity 2.2: Implement the Policy Intervention Simulator

Enable users to select real-world policy interventions and instantly see the projected effect on the baseline HDI score, using boost multipliers defined per intervention in data.ts.

```
// data.ts — policy intervention definitions
export const POLICY_INTERVENTIONS: Policy[] = [
  {
    category: 'Health',
    title: 'National Primary Healthcare Revitalization and Sanitation Drive',
    estimatedBoost: 0.035,
    difficulty: 'High',
    timeline: 'Medium-Term',
  },
  // ... Education, Income interventions
];

// App.tsx — applying selected interventions
function getSimulatedValues(baseline: HdiComputation, selected: Policy[]): HdiComputation {
  const totalBoost = selected.reduce((sum, p) => sum + p.estimatedBoost, 0);
  return { ...baseline, hdiScore: Math.min(1, baseline.hdiScore + totalBoost) };
}
```

Milestone 3: Backend & AI Advisor Development

Milestone 3 focuses on building the Express backend that powers the AI Policy Advisor feature — the component of server.ts that receives a country's indicator profile and returns a Gemini-generated advisory report.

Activity 3.1: Writing the /api/advisor Route Handler

```
// server.ts — AI Policy Advisor endpoint
app.post('/api/advisor', async (req, res) => {
  const { hdiScore, category, healthIndex, educationIndex, incomeIndex } = req.body;

  if (!hdiScore || !category) {
    return res.status(400).json({ error: 'HDI score and category are required.' });
  }

  try {
    const prompt = buildAdvisorPrompt(hdiScore, category, {
      healthIndex, educationIndex, incomeIndex
    });
    const result = await geminiClient.generateContent(prompt);
    const advisory = extractStructuredAdvisory(result.response.text());
    res.json({ success: true, advisory });
  } catch (err) {
    res.status(500).json({ error: 'Failed to generate policy advisory.' });
  }
});
```

Define Prompt Engineering Helpers:

- **buildAdvisorPrompt():** Assembles a structured prompt combining the HDI category, sub-index breakdown, and instructions to identify the weakest development dimension.
- **Diagnostic instructions:** The prompt explicitly asks Gemini to detect 'strengths' and 'weaknesses/gaps' — for example, flagging substantial longevity deficits when the health index is the lowest of the three.
- **Recommendation instructions:** The prompt requests categorized reform recommendations (Health, Education, Income) each with an estimated HDI score boost, a difficulty rating, and an implementation timeline.

- **extractStructuredAdvisory():** Parses the Gemini response into strengths, weaknesses, and a list of recommendation objects returned to the frontend.

Milestone 4: Frontend Development

Milestone 4 focuses on building a polished, dark-themed interface for the HDI Predictor. This involves designing a responsive two-panel layout with React and TypeScript, and wiring up dynamic state management so slider changes, scenario selection, and AI advisory requests all update the UI instantly.

Activity 4.1: Designing the User Interface

- **Set Up the Base Layout:** Build the main App.tsx shell with a hero section, a two-column input/output layout, and a tabbed navigation bar (Predictor, Comparison, Report, About).
- **Design a Dark, Data-Dense Theme:** Implement a navy/indigo colour palette with card-based panels for inputs, prediction results, radar charts, and diagnostics.
- **Create Separate UI Components:** HdiPredictor.tsx (sliders + preset dropdown), a Prediction Result card (HDI score gauge, confidence margin, sub-index breakdown), an Indicator Contribution Radar chart, and a Sovereign Diagnostics panel (strengths/weaknesses).

Activity 4.2: Creating Dynamic Content Rendering

- **Handle Slider Input:** Attach onChange handlers to each of the four indicator sliders, triggering an instant recalculation via calculateHdiFromIndicators().
- **Render the Prediction Result:** Display the HDI score in a circular gauge, the tier badge (e.g. 'MEDIUM HDI'), the confidence margin, and the three sub-index values (Health, Education, Standard of Living).
- **Render the Recommendation Cards:** After a 'Consult AI Advisor' request, dynamically render categorized reform cards (Health, Education, Income) each showing the estimated boost, difficulty, and timeline returned by the backend.

Milestone 5: Deployment

In Milestone 5, the focus is on deploying the HDI Predictor application — first locally to verify correct operation of both the client-side computation engine and the Gemini-powered advisor endpoint, then publicly to a hosting platform for evaluator and stakeholder access.

Activity 5.1: Preparing the Application for Local Deployment

```
D:\projects>npm install
D:\projects>cp .env.example .env.local
D:\projects>echo GEMINI_API_KEY=your_gemini_api_key_here >> .env.local
```

Activity 5.2: Local Testing and Verification

```
D:\projects>npm run dev
  VITE v5.x.x ready in 320 ms
  ➜ Local: http://localhost:5173/
  ➜ Network: use --host to expose
```

Once running, open a browser and navigate to http://localhost:5173 to access the app. Interact with the Human Capital Predictor sliders, load a country preset, run the Policy Intervention Simulator, and click "Consult AI Advisor" to verify the Gemini-generated recommendations are correctly returned and rendered.

Activity 5.3: Public Deployment

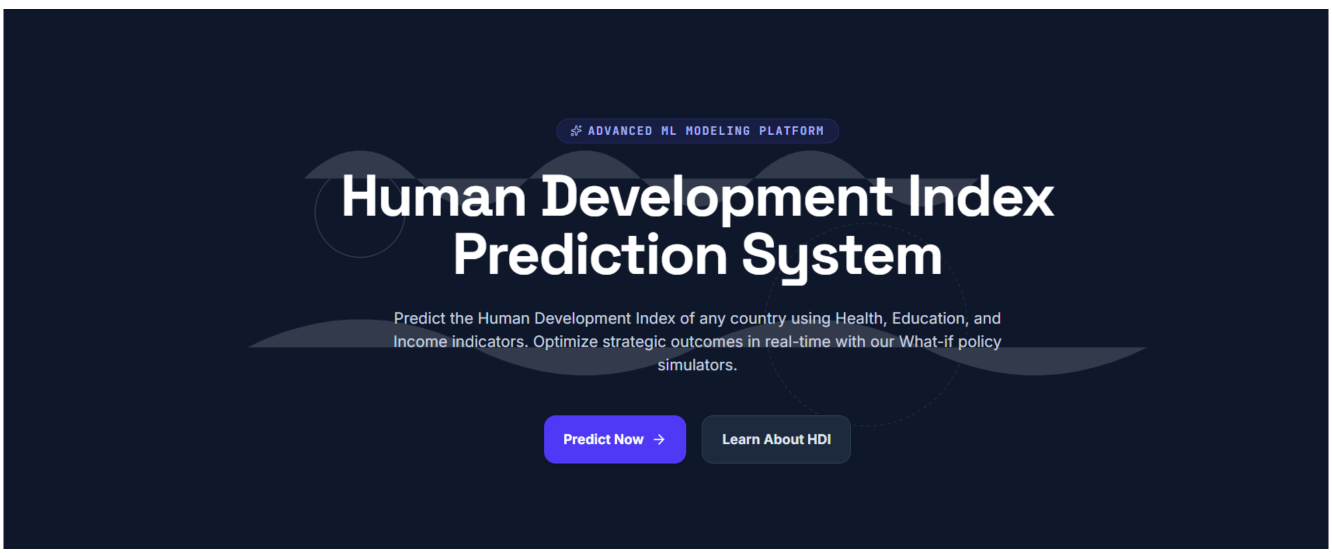
- Deploy the static frontend build to Vercel or Netlify for global CDN-backed hosting.
- Deploy the Express /api/advisor backend as a Node.js server (Render, Railway) or as a serverless function alongside the frontend.
- Set the GEMINI_API_KEY as a secret environment variable on the hosting platform — never commit it to the repository.

Conclusion:

HDI Predictor — Policy Reviewer is a real-time development analytics platform built with React, TypeScript, and an Express.js backend integrated with Google's Gemini API. It instantly computes a country's Human Development Index using the UNDP geometric mean methodology, simulates the projected impact of real-world policy interventions, benchmarks custom profiles against global country data, and generates AI-powered administrative reform recommendations tailored to each country's specific development gaps. The project, which included model definition, computation engine development, AI advisor integration, and deployment, demonstrates how combining deterministic statistical modelling with generative AI can produce actionable, evidence-based policy guidance. Looking ahead, the platform offers significant opportunities for expansion — such as historical HDI trend tracking, multi-country side-by-side simulation, and direct integration with the live UNDP Human Development Report API — positioning it to evolve from a single-session analytical tool into a comprehensive sovereign development planning assistant.

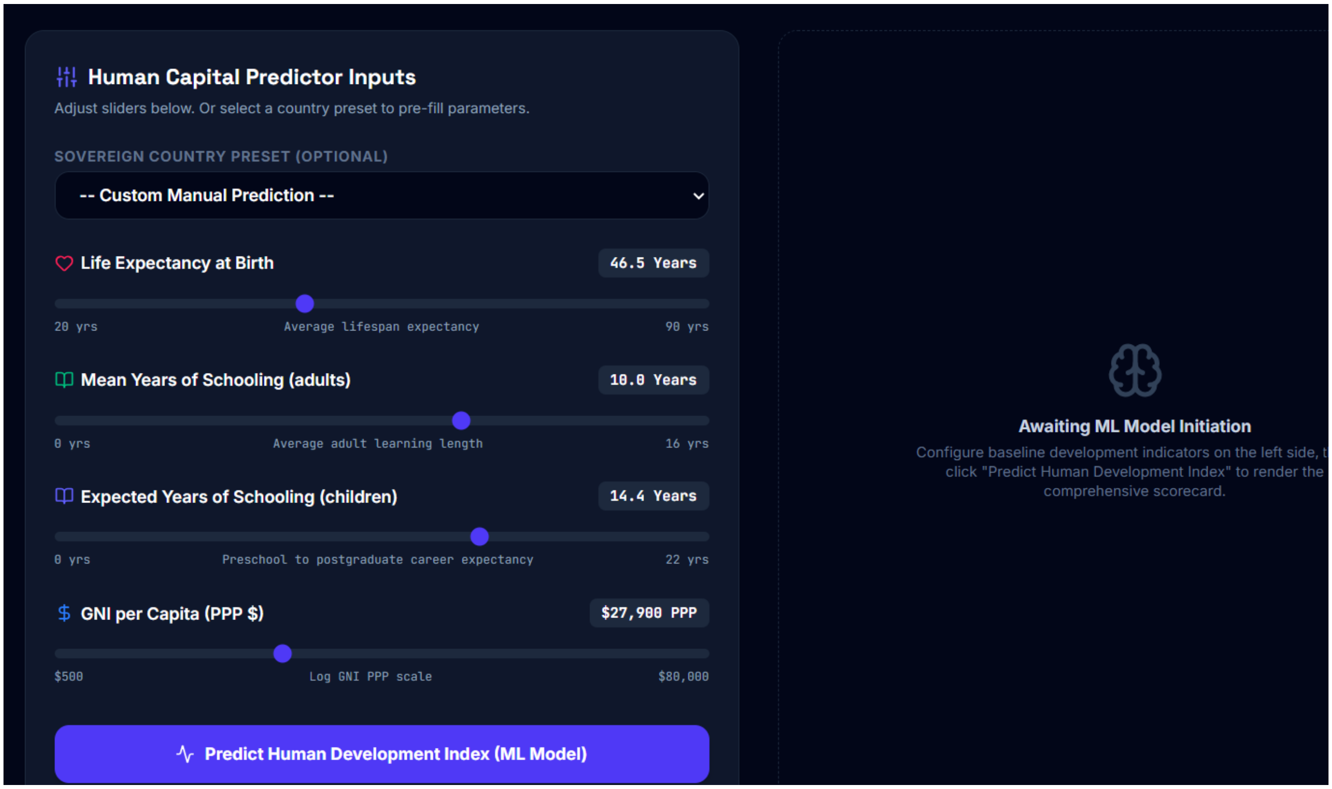
Exploring the Website's Web Pages:

Home / Hero Page:



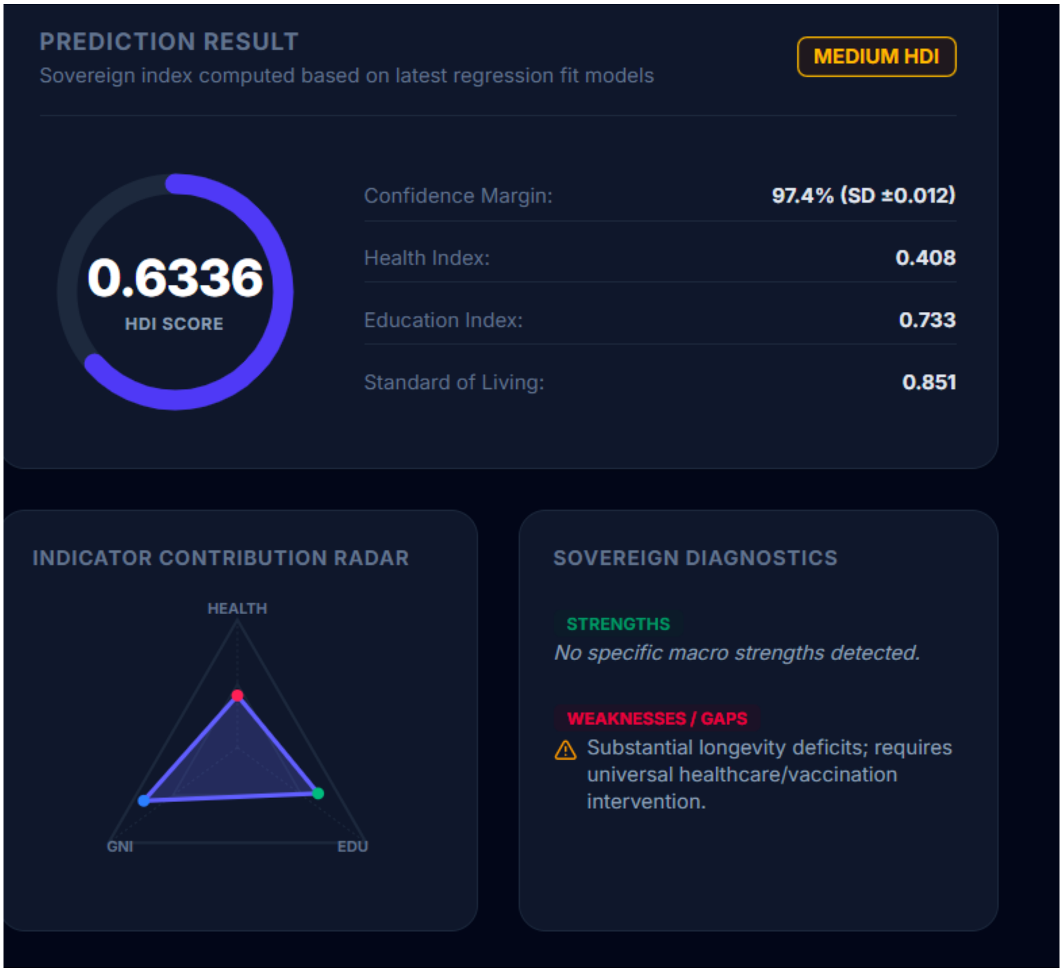
Description: The landing section of HDI Predictor introduces the platform as an "Advanced ML Modeling Platform", with a bold headline framing the tool's purpose — predicting the Human Development Index of any country using Health, Education, and Income indicators. Two primary calls to action, "Predict Now" and "Learn About HDI", guide users directly into the predictor tool or the educational methodology page. The dark navy background with subtle wave graphics establishes a professional, data-driven visual identity.

Human Capital Predictor Inputs:



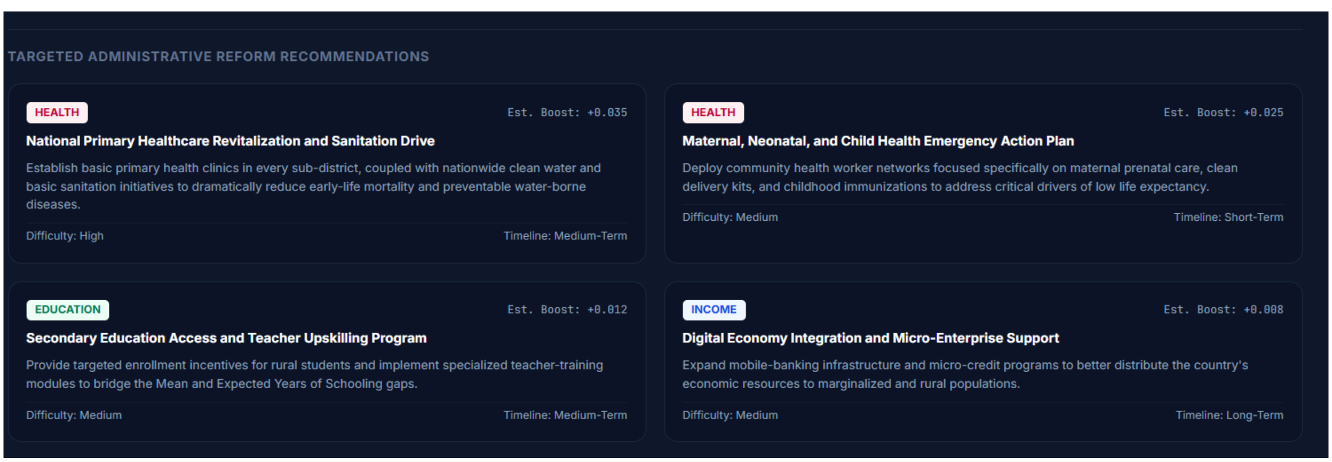
Description: The input panel allows users to select a sovereign country preset from a dropdown or manually adjust four indicator sliders: Life Expectancy at Birth (20–90 years), Mean Years of Schooling for adults (0–16 years), Expected Years of Schooling for children (0–22 years), and GNI per Capita on a logarithmic PPP scale ($500–$80,000). Each slider displays the current value alongside descriptive context (e.g. "Average lifespan expectancy"). The right panel remains in an "Awaiting ML Model Initiation" state until the user clicks "Predict Human Development Index".

Prediction Result Panel:



Description: Once a prediction is generated, this panel displays the composite HDI Score (0.6336) in a circular gauge alongside a tier badge ("MEDIUM HDI"), a model confidence margin (97.4% ± 0.012), and the three sub-index values — Health Index, Education Index, and Standard of Living. Below, an Indicator Contribution Radar chart visualizes the relative strength of each dimension, while the Sovereign Diagnostics card automatically flags weaknesses — in this example, identifying "substantial longevity deficits" and recommending universal healthcare and vaccination intervention.

Targeted Administrative Reform Recommendations:



Description: Generated by the AI Policy Advisor, this section presents categorized reform recommendations as individual cards — colour-coded by dimension (Health, Education, Income). Each card includes the estimated HDI score boost (e.g. +0.035 for a National Primary Healthcare Revitalization initiative), a plain-language description of the intervention, a difficulty rating (Low/Medium/High), and an implementation timeline (Short-Term/Medium-Term/Long-Term) — giving policymakers a prioritized, quantified action plan for improving the country's Human Development Index.